Manipulation-aware Active Perception for Aerial-Ground Cooperative Mobile Manipulation

Research manuscript first draft

Abstract

Aerial sensing can reveal regions that a ground robot cannot readily inspect, but maximizing environmental information alone need not reveal the locations that make a manipulation task executable. We study this mismatch in cooperative retrieval by an aerial robot and a ground mobile manipulator. Our framework projects validated inverse-reachability candidates through the ground robot's footprint to identify environment cells relevant to future manipulation. It combines this support with an observation-deficit field and a finite-window LiDAR acquisition model to select aerial viewpoints. A shared execution interface requires measured ground support, geometric compatibility and collision-aware manipulation screening before handoff; sensing predictions never count as observations. We evaluate the complete perception-to-retrieval loop on a P450–BUNKER–AUBO i5–AG95 physics-based simulation platform using 96 independent scenes and 192 primary paired trials. Under the same three-window observation budget, manipulation-aware selection achieves physical retrieval in 80/96 scenes, compared with 56/96 for a generic NBV policy that differs only in gain weighting. The paired difference is 25.00 percentage points, with a prespecified conservative 95% confidence interval of [8.56, 39.15] and an exact McNemar $p = 8.43 \times 10^{-6}$. These results support task-conditioned sensing for the studied static, known-object retrieval population; they do not establish a general time-saving or real-robot performance claim.

1 Introduction

An aerial robot and a ground mobile manipulator offer complementary capabilities: the former can change its observation position without following a ground route, whereas the latter can support a manipulator and physically retrieve an object. Exploiting this complementarity requires more than detecting a target from the air. The ground robot must eventually occupy a suitable base pose, see the target at close range, and execute a collision-aware grasp and lift. Information that improves a map is therefore not necessarily information that makes the retrieval task executable.

Next-best-view (NBV) planning provides a natural mechanism for deciding where to observe. Exploration and reconstruction approaches select measurements that reveal unmapped space or incompletely observed surfaces [1, 2]. Such objectives address their intended mapping tasks, but do not by themselves express the capabilities of a different robot that will act later. For retrieval, a large poorly observed area outside useful manipulator base placements can contribute substantial generic observation gain while providing little evidence about an executable manipulation pose. Conversely, a smaller unresolved region intersecting a promising base footprint may prevent the entire task from proceeding. With a finite sensing budget, this distinction makes the spatial allocation of observations consequential.

Task-oriented view selection is not new: prior work selects views for grasp detection or decides when to stop observing and grasp [4, 3]. We address a different coupling. The aerial sensing robot does not execute the grasp, and the region that needs observation is not restricted to the object's visible surface. It depends on where the ground manipulator can stand, the area occupied by its base, and the operational constraints between that stance and a completed retrieval. The research question is whether these downstream manipulation requirements can guide aerial observation more effectively than a shared, information-centric NBV objective under the same observation budget.

Our central idea is to propagate manipulation support backward into the sensing objective. Given an aerially perceived grasp target, an inverse-reachability query produces candidate base poses. Validated candidates supply a joint-margin-based manipulation relevance field. Their exact base footprints project this relevance into environment space, where it weights the current observation deficit. A finite-scan acquisition model then estimates which cells a candidate aerial view can observe during the next accepted window. The policy selects a view using the resulting task-weighted gain and a shared motion-cost proxy.

After each observation, the system rechecks whether an exact base candidate is sufficiently observed and passes a common execution screen. This connects observation choice to a physical handoff rather than treating a heatmap or a reachable grasp as the final outcome.

We make three contributions:

1. A manipulation-aware active-perception framework for heterogeneous aerial-ground retrieval, in which the ground robot's future operation informs the aerial robot's sensing decisions.

2. A footprint-aware task relevance formulation that combines validated manipulator reachability, exact base-pose support and operational geometry with environmental observation deficit, while keeping unsupported capability, unobserved ground and observed collision evidence distinct.

3. Closed-loop physics-based validation on P450, BUNKER, AUBO i5 and AG95, using 96 independent scenes and 192 primary paired trials with physical retrieval success as the primary endpoint. A controlled Generic/Ours comparison shares the sensing model, candidates, costs, budget and execution pipeline, isolating the observation-gain weighting.

The proposed contribution is neither a new inverse-kinematics solver nor the shared platform corrections. The evaluation supports a success-rate advantage within the defined simulated task population; efficiency, failure mechanisms and transfer limitations are reported separately.

2 Related Work

NBV and active scene acquisition. Bircher et al. select exploration motions through a receding-horizon NBV planner whose branches are valued by newly observable space [1]. Surface Edge Explorer instead uses the density and boundaries of existing 3D observations to direct further surface acquisition [2]. These approaches illustrate different representations for deciding where additional measurements are useful. Our work does not argue that generic exploration objectives are incorrect for mapping. It asks whether their spatial weighting is appropriate when the endpoint is another robot's manipulation task. Accordingly, our Generic comparator is an objective-level control in the same implemented planner, not a claimed reproduction or head-to-head evaluation of these complete published systems.

Task-oriented perception and active grasping. Gualtieri and Platt study how viewpoint selection affects the localization and confidence of grasp detections [4]. Breyer et al. combine exploration of occluded target regions with continuously updated grasp predictions and a decision to execute or continue observing [3]. These works establish that the value of a measurement can depend on a grasping objective rather than only on map completeness. Our formulation transfers this reasoning across heterogeneous robots: aerial observations are weighted by the ground mobile manipulator's feasible stance support, including its extended base footprint. The relevance of a ground cell is thus derived from future base placements, not solely from whether it belongs to an incompletely observed target. We do not claim to introduce task-oriented perception in general.

Aerial-ground cooperation. ColAG couples aerial perception and mapping to the navigation needs of perception-limited ground vehicles, including aerial waypoint scheduling [6]. Lenz et al. demonstrate autonomous perception and brick handling with ground and aerial systems for wall construction [5]. These studies motivate treating cooperative sensing and physical action as an integrated robotics problem. Our emphasis is the intermediate representation linking a ground arm's manipulation support to aerial viewpoint value, and its evaluation against a matched generic objective through physical retrieval. We do not claim general multi-robot task allocation or a replacement for collaborative navigation and localization.

Reachability and mobile manipulation. Capability-map work represents the structure of a robot's reachable workspace [8]. RM4D provides a compact representation supporting both reachability and inverse-reachability queries, including mobile-manipulator base-placement use cases [7]. We reuse RM4D as the source of candidate manipulation support rather than presenting it as a contribution of this paper. The additional step is to project validated base-pose support into the environment that must be sensed before those poses can be used. Reachability alone remains insufficient: base occupancy, measured ground support, close-range visibility and collision-aware execution are evaluated downstream. Our shared execution interface makes these limitations visible in the experimental outcome instead of assuming that a valid inverse-kinematics solution completes the task.

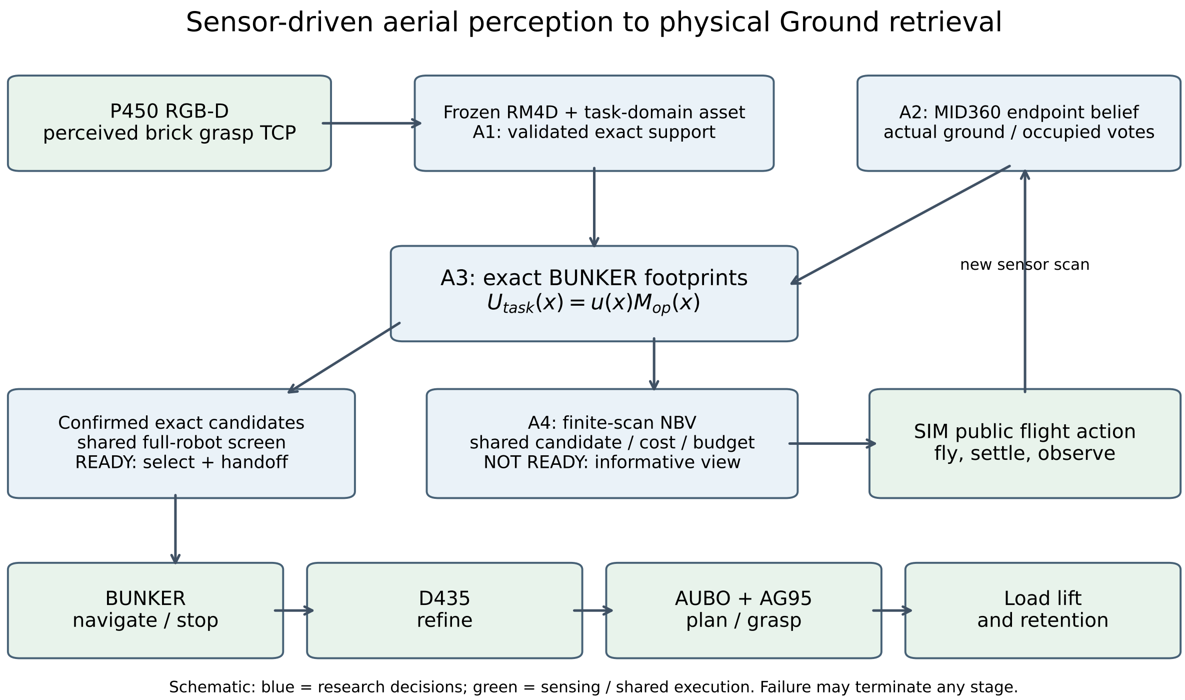

Figure 1: System overview. Manipulation support and measured environment evidence guide aerial sensing, while a common execution interface connects the selected exact base pose to physical retrieval. Schematic, not a trial reconstruction.

3 Method

3.1 Problem and system overview

Let $\widehat{T}_g \in SE(3)$ be a grasp TCP pose estimated from aerial RGB-D observations in a shared map frame. A ground base pose is $q = (p_x, p_y, \psi) \in SE(2)$, with p denoting the BUNKER base-link ground-plane position and ψ its yaw. An aerial viewpoint is $v = (p_v, \psi_v)$ under the implemented level-hover assumption. The task is to select observations and an executable ground stance that lead to physical retrieval within a limited number of accepted sensing windows. We do not solve an optimal joint belief-space control problem; we use an interpretable one-step sensing surrogate followed by explicit execution checks.

Figure 1 shows the loop. Aerial RGB-D initializes the target; inverse reachability supplies ground-pose support; measured LiDAR endpoints update environmental evidence; and a task-weighted viewpoint policy requests further sensing unless a confirmed candidate passes the common execution screen. The ground robot then navigates, refines the grasp from wrist RGB-D, and performs the physical operation. All environment fields use aligned 0.10 m cells in map coordinates. Algorithms consume sensor observations and public poses/TF, not scene-generator target coordinates. The simulation-specific localization and contact backends are disclosed in Section 6.2.

3.2 Validated manipulation relevance

An RM4D inverse query [7] produces base candidates $\{q_i\}$ for $\widehat{T}_g$. The query uses an explicit rigid frame bridge between the public map and the calibrated robot-reference convention; returned poses are transformed back to map. The perception target itself is not shifted for subsequent manipulation. A separate Ground-task map supplies the required workspace coverage without modifying the reference RM4D algorithm.

For each evaluated candidate, define a validity indicator g_i. It is one only when reachability, inverse kinematics, the validator's collision checks and minimum joint-limit margin all pass. For the associated joint solution θ_i, let $m_i = \min_j \{\theta_{ij} - \theta_j^{\min}, \theta_j^{\max} - \theta_{ij}\}$. The continuous quality is

$$r_i = \begin{cases} \mathrm{clip}(m_i/m_{\mathrm{sat}}, 0, 1), & g_i = 1, \\ 0, & g_i = 0. \end{cases} \tag{1}$$

The gate requires $m_i \geq m_{\min}$ when a valid joint solution exists; invalid candidates need not have a defined margin. The executed settings use $m_{\min} = 0.01$ rad and $m_{\mathrm{sat}} = 0.5$ rad. Numerical IK/FK residuals are acceptance diagnostics, not manipulation quality. Travel distance and the baseline's final ranking score do not enter r_i.

For a base-grid cell C, choose the exact validated winner

$$i_C^* = \arg \max_{i:p_i \in C,\, g_i = 1} r_i, \qquad q_C^* = q_{i_C^*}, \qquad R(C) = r_{i_C^*}. \tag{2}$$

Ties retain the first candidate in the fixed evaluated order, independently of environment evidence. Geometry is anchored at q_C^*, not at the grid-cell center. This avoids introducing an unvalidated displacement merely to rasterize support. Non-winning candidates are diagnostic and are not reselected as support because an environment happens to favor them.

The planner validates at most 256 candidates. A cell containing no evaluated candidate remains *unassessed*; a cell containing evaluations but no valid one has no validated feasible support in that evaluated set. Neither statement proves complete continuous-space infeasibility. Thus R is a validated, partially assessed interest field, not a complete capability map. No smoothing fills gaps or spreads support to unassessed base poses.

3.3 Environment evidence and operational footprints

Each accepted LiDAR window supplies endpoints transformed into map using the sensor-time public TF. For a horizontal ground plane z_0, a usable endpoint within the ground tolerance supplies ground evidence; one at or above the obstacle-height threshold supplies occupied evidence. The executed tolerances are 0.02 m and 0.05 m, respectively. Returns in the intermediate height band do not create either vote. Evidence is accumulated per cell per window, not per individual point. Raw occupancy prioritizes an occupied vote over a free vote in the same window, and historical occupied evidence is not erased by later ground returns. Two free-vote windows are required for the raw FREE label when no occupied evidence exists; other cells are UNKNOWN. Aerial rays are not carved as entire free XY segments: a ray can pass above an obstacle without observing the ground underneath it.

Let $N_t(x)$ count informative raw evidence windows for cell x. The observation-deficit score is

$$u_t(x) = \exp[-N_t(x)/\tau], \qquad \tau = 2. \tag{3}$$

This is a heuristic, not a calibrated probability or entropy. In particular, FREE does not force $u_t = 0$, and a low deficit does not certify collision freedom.

Let $F(q)$ be the padded continuous rectangular base footprint and

$$K(q) = \{x : \mathrm{cell}(x) \cap F(q) \neq \varnothing\}. \tag{4}$$

The padded half-length and half-width are 0.52 and 0.39 m. Closed cell/rectangle overlap includes boundary contact, rather than testing only whether cell centers lie inside the footprint.

Operational evidence distinguishes measured endpoints associated with the known target, environmental obstacles, and ambiguous returns. Target association uses runtime perception and geometric consistency, not scene truth. The target is checked as a perceived, allowance-expanded continuous object against $F(q)$; a shared coarse cell alone is insufficient to declare target/base collision. Environmental occupied cells remain conservative blockers. Complete ambiguous endpoint records are checked using their declared local uncertainty disks against $F(q)$; missing endpoint history retains the conservative cell blocker. The executed disk radius is 0.033 m, an engineering allowance conditional on the public-map TF profile, not a calibrated physical error bound. Ambiguous returns are not converted to target or free evidence. The target association allowance, endpoint uncertainty and execution clearance are distinct declared quantities, not interchangeable safety guarantees.

Denote this operational blocker by $B_t(q) \in \{0, 1\}$. Separately, let $H_t(x)$ count windows containing an actual ground endpoint in x, even if the same cell also contains an occupied endpoint elsewhere. Confirmation requires

$$A_t(q) = \mathbf{1}[B_t(q) = 0]\, \mathbf{1}[F(q) \text{ is inside the grid}] \prod_{x \in K(q)} \mathbf{1}[H_t(x) \geq 2]. \tag{5}$$

Keeping H_t separate preserves measured ground presence without clearing occupied evidence. Unknown ground does not block a pose geometrically, but it does prevent confirmation. $B_t = 1$ means blocked by this operational model, not a proof of global navigation infeasibility. The same assessment is used for relevance support and exact-candidate handoff.

3.4 Task relevance in environment space

The nominal and operational support projections are

$$M_{\mathrm{nom}}(x) = \max_{C: x \in K(q_C^*)} R(C), \tag{6}$$

$$M_{\mathrm{op},t}(x) = \max_{C: x \in K(q_C^*),\, B_t(q_C^*)=0} R(C). \tag{7}$$

Only validated winners participate. Where nominal support exists but all covering poses are blocked, operational support is zero. Where no nominal support exists, the implementation retains a distinct no-validated-support state and excludes that cell from the task sum; it does not reinterpret the cell as observed or physically unreachable. The task field is

$$U_{\mathrm{task},t}(x) = u_t(x) M_{\mathrm{op},t}(x). \tag{8}$$

The maximum avoids counting one environment cell repeatedly simply because several useful base poses cover it. This formulation retains unknown regions that could support a useful manipulation stance while excluding already blocked stances. Occupancy does not overwrite $R(C)$. Grid-clipped footprints may contribute to the in-grid projection but cannot be confirmed by Eq. (5). Both manipulation coverage and environment uncertainty remain explicit.

3.5 Finite-window task-aware viewpoint selection

A cell being inside the LiDAR field of view does not guarantee that a finite scan window measures it. We therefore use the installed cyclic scan directions and publisher angular/discard rules to predict ground intersections. For each scheduled packet-start phase $s \in \mathcal{S}$, define $h_s(v, x)$ as one if at least one emitted ray during the next five-second window reaches cell x on the nominal ground plane, satisfies range limits, and is not occluded. The acquisition opportunity is

$$w_t(v, x) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} h_s(v, x). \tag{9}$$

This uniformly averages scheduled starting phases; it is not an empirically calibrated return probability. Environmental occupied cells and incomplete ambiguous evidence form assumed-height prisms; complete ambiguous endpoints form local cylinders; the perceived target uses its oriented box. The 1 m prism/cylinder height is an occlusion surrogate, not an estimated obstacle height. Unknown space does not occlude. Segment tests distinguish intersection from passing above an occluder. These predictions never increment evidence counts, and unmodeled interception or pose/scan mismatch can still cause missed returns.

One additional informative window would reduce Eq. (3) by

$$\Delta u_t(x) = (1 - e^{-1/\tau}) u_t(x) = \alpha u_t(x). \tag{10}$$

Holding the current manipulation support fixed for one-step prediction gives

$$G_{\mathrm{task}}(v) = \alpha \sum_x w_t(v, x) U_{\mathrm{task},t}(x), \tag{11}$$

$$G_{\mathrm{generic}}(v) = \alpha \sum_x w_t(v, x) u_t(x). \tag{12}$$

Equation (11) is a predicted observation-gain surrogate: it favors observable task-relevant deficit mass, not calibrated expected information gain, guaranteed confirmation, or the probability of task completion. The current field is recomputed after actual observations.

Both methods use

$$S_m(v) = G_m(v) - \lambda C(v), \quad C(v) = \|p_v - p_{\mathrm{cur}}\|_2 + \kappa |\operatorname{wrap}(\psi_v - \psi_{\mathrm{cur}})|. \tag{13}$$

The cost is a displacement/yaw proxy, not a measured path length or energy model. The shared settings are $\lambda = 0.25$ and $\kappa = 0.25\,\mathrm{m/rad}$. Candidates include the current pose and a task-independent local XY lattice with offsets $\{-2, -1, 0, 1, 2\}$ m, fixed current altitude and yaw alternatives, filtered by common bounds. The tilted LiDAR mount makes yaw relevant despite its panoramic horizontal field of view. Ties use cost then fixed candidate index. Figure 2 shows the controlled distinction: only G_m's manipulation weighting differs.

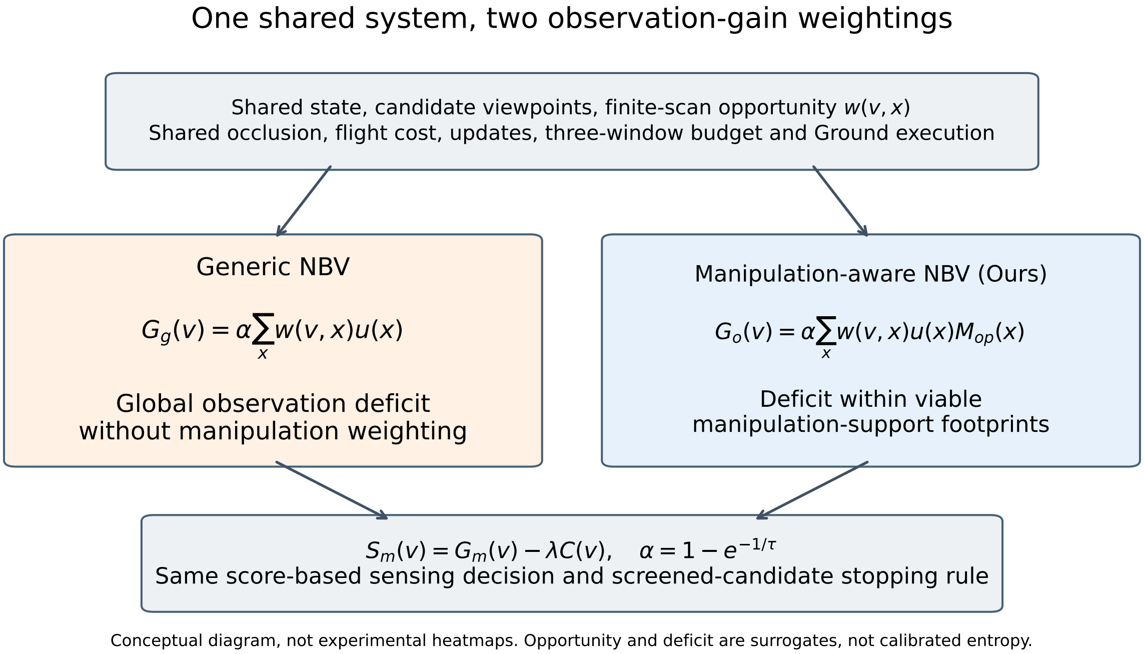

Figure 2: Matched NBV policies. The finite-scan opportunity, candidate set, motion cost, evidence updates and execution rules are shared. Manipulation support weights the deficit only in Ours. The diagram is conceptual, not a new result.

3.6 Ground execution interface and stopping

After each accepted update, exact winners satisfying Eq. (5) are considered in the fixed relevance order. The common execution layer screens at most four confirmed candidates using the complete base, arm, gripper, target and post-grasp load models, with bounded observation-pose, IK and approach searches. A successful screen terminates active perception and supplies the exact candidate for navigation; the system does not use the field-cell center or a previously successful stance. If no screen succeeds, the policy requests its highest-scoring admissible view while budget and positive score permit. Nonpositive score, no usable view or budget exhaustion can terminate without handoff. The initial window counts toward the three-window limit.

Screening is only a preview. BUNKER must actually arrive and stop, wrist D435 must provide a fresh usable target estimate, and the arm must establish a collision-aware refined pregrasp at the measured arrival pose. We call this executed discovery stage D_{exec}. Descent, physical closure, lifting and retention follow with their own checks. The common 12 mm development clearance and SIM-specific velocity-feedback mode are execution settings shared by all applicable comparisons, not method-specific contributions. A reachable target does not guarantee a usable camera view or a complete collision-free operation. Failure at any of these stages remains a task failure, not a reason to redefine success.

4 Experiments

4.1 Platform and task

We evaluate the fixed `paper1-eval-finite-scan-v1` system in Gazebo physics-based simulation. P450 supplies aerial RGB-D and MID360 observations; BUNKER carries an AUBO i5 arm, an AG95 gripper and a wrist-mounted D435 camera. The shared stack uses PX4 SITL/Prometheus for aerial control, ROS navigation for the base, and MoveIt-based arm planning. Each task begins with normal aerial target perception and proceeds through the policy's own observations, base selection and robot execution. No successful base pose is manually assigned. The perceived target is a known brick on static horizontal ground. Physical grasping uses simulated contact and normal controller execution, not an object teleportation or a substituted success label.

All methods use the same initial observation pose $(x, y, z, \psi) = (-1.4, 0, 1.2, 0)$ in map, the same RGB-D acceptance gate, aligned 0.10 m fields, and the settings of Section 3 where applicable. The actual collection revisions are AGENT `3d13a95`, SIM `a0ae8e3` and original RM4D `e9d4312`, with the separately calibrated Ground-task asset. These identify the executed system rather than assuming that a later development tip reproduces it. No robot decision rule, scene seed, sensor gate or physical success predicate changed during collection.

4.2 Independent scenes and paired protocol

The primary study contains 96 independently sampled scenes. Difficulty is drawn independently with probability 1/3 for Easy, Moderate and Hard, giving the prospectively fixed realization of 38, 28 and 30 scenes. Easy contains no wall, Moderate one wall, and Hard two walls. The walls are 1 m high; the Moderate wall is 1 m long and Hard walls 0.6 m long, all with 0.15 m thickness. Target position/yaw, initial base pose, and wall position/yaw are independently perturbed according to the frozen generation rules. The target is centered near $(2, 0)$ and the base near $(3, -2.5)$ m. Target XY perturbations are bounded by 0.15 m and yaw by $\pi/6$; wall-center perturbations are bounded by 0.15 m and wall yaw by $\pi/12$.

The Hard generator introduces opportunities for occlusion alongside unknown space that need not support manipulation; it does not guarantee that each realization produces either a blocked useful region or a favorable method difference. Seeds were disjoint from development data, and the complete geometry, method order and auxiliary subsets were fixed before method runs. Admissibility checks concern initial geometry and sensor setup, not candidate counts, path existence, viewpoint scores or retrieval outcomes. Each scene is run once per primary method in the predeclared balanced/shuffled order, yielding 192 planned primary trials. The paired scene, not the view or candidate, is the independent analysis unit. We retain the realized mixture rather than post-stratifying tiers after observing their outcomes.

4.3 Comparators and ablations

The sole primary contrast is *Generic NBV* versus *Ours*. Generic uses Eq. (12); Ours uses Eq. (11). They share initial conditions, candidate generation, finite-scan opportunity and occlusion, cost, observation budget, evidence updates, exact support, operational gating, candidate screening and the full execution pipeline. Their later measurements may differ because their actions differ. Generic is also the without-task-weighting ablation; it is not counted a second time.

Two context controls run on six predefined scenes, the first two generated IDs in each tier. *RM4D-only* uses the initial aerial RGB-D target and calibrated task-domain map, selects the baseline top candidate, and skips MID360 confirmation and pre-handoff multi-candidate screening. Downstream physical checks remain enabled. It is therefore a context control, not a controlled isolation of relevance weighting or a test of the original literature map's domain. *Fixed-view* reobserves the current measured hover location without selecting translated NBVs, retaining the three-window cap and shared screening stop.

On the first four generated Hard IDs, *without occlusion* removes occlusion only from the predicted gain, not from collision blocking or execution checks; *without flight cost* sets the policy flight weight to zero. The scheduled Ours counterpart is reused for each auxiliary comparison. These 12 context and 8 ablation tasks bring the total to 212 planned tasks, not 212 primary trials. The small auxiliary subsets provide descriptive context and do not support powered secondary superiority, equivalence or universal-necessity claims.

4.4 Outcomes, failures and observation cost

The primary binary endpoint is physical retrieval success. It requires the perception, arrival, grasp and lift/short-retention checks, including an independent simulated-object height check. The existing checker requires both TCP lift and measured object-height increase of at least 0.10 m; the task also requires its existing one-second wall-clock hold of fresh grasp confirmation. This interface hold is distinct from the simulation-time efficiency metrics below. Candidate confirmation, a successful planning preview, D_{exec}, TCP motion or a task-process success message alone does not establish retrieval. We record progress through observation, confirmation, screening, navigation/stopping, D435 refinement, D_{exec}, grasp and lift/retention. First-terminal failure counts assign one cause to an unsuccessful task; downstream stages that were not reached do not create additional failures. Model rejection is not interpreted as proof that a physical collision occurred.

The common observation budget is three accepted five-simulation-second windows, including the initial window. A further observation may follow translation, yaw-only motion or reobservation near the same location. We distinguish accepted windows from actual changes between measured observation poses. A

Table 1: Primary physical retrieval success under the shared observation budget. The table is directly included from the existing evaluation asset.

Method	Retrieval success	N	Percent
Ours	80	96	83.33
Generic	56	96	58.33

resolved location change exceeds 0.20 m translation; wrapped yaw change exceeding 0.20 rad is separately identifiable. These are reporting categories, not flight-path lengths or additional control thresholds.

Robot-efficiency times use simulation time. Active time starts at the first capture and ends at active stop or terminal failure, including observation, motion, waiting, computation and shared previews. It excludes initial takeoff, aerial RGB-D and the initial reachability query. Ground time starts at navigation and ends at the task's terminal Ground event. Whole-task time includes initialization and the normal return/landing sequence. Unreached Ground stages have missing Ground time, not zero cost. Distance traces are incomplete in this release, so distance saving is not an inferential endpoint or claim.

Normal insufficient sensing, bounded planning rejection and physical execution failure remain valid outcomes. Only documented infrastructure-invalid attempts may receive one same-scene/method replacement within the fixed reserve. All 212 planned slots have valid selected binary outcomes and all 96 primary pairs are complete. Two proven-invalid originals received replacements, giving 214 formal activations; one additional external bare simulator startup was separately charged to resources. Original attempts are preserved. These activations are not additional independent samples.

4.5 Prespecified analysis

Let a, b, c, d denote both-success, Ours-only, Generic-only and both-failure paired counts. We report the marginal success rates and

$$\widehat{\Delta} = (b - c)/96. \tag{14}$$

The single primary test is the two-sided exact McNemar test, conditioning on $b + c$ discordances, with significance level 0.05. The prespecified conservative 95% interval for Δ combines two 97.5% Clopper–Pearson intervals $[L_b, U_b]$ and $[L_c, U_c]$ for $b/96$ and $c/96$ as $[L_b - U_c, U_b - L_c]$. Bonferroni joint coverage is at least 95%; independence of the two discordant categories is not assumed. The interval concerns the paired risk difference, not the test's p-value. A first-activation sensitivity counts infrastructure-invalid originals as unsuccessful without dropping denominators.

Tier-specific outcomes and auxiliary comparisons are descriptive, not additional primary tests. Conditional efficiency is summarized on the 53 scenes where both primary methods retrieve successfully, with 10,000 whole-scene paired bootstrap resamples and the fixed analysis seed. Conditioning is explicit: these means are not unbiased all-task cost effects, and a failed quick task is not efficient retrieval. The existing analysis also reports all-96-denominator success-resource curves. No additional experiment, significance test or outcome-driven sample extension is introduced in this manuscript.

5 Results

5.1 Primary physical retrieval outcome

Ours retrieves successfully in 80 of 96 scenes (83.33%), compared with 56 of 96 (58.33%) for Generic (Table 1). There are 27 Ours-only and 3 Generic-only successes, alongside 53 joint successes and 13 joint failures (Table 2). The paired difference is +25.00 percentage points, with the prespecified conservative 95% interval $[+8.56, +39.15]$ percentage points and exact McNemar $p = 8.43 \times 10^{-6}$ (Table 3). This supports a success advantage in the specified simulated scene population under the common budget. Because the primary policies differ only in gain weighting, common platform and execution improvements cannot explain a method-exclusive implementation advantage within this comparison. The result does not compare against every published NBV planner or establish performance outside the sampled population.

The three Generic-only scenes remain in the analysis. Two involve an Ours perceived target/base collision rejection and one loss of grasp confirmation during lift/retention. In the first-activation-invalid-as-failure sensitivity, the paired counts become 52/27/4/13, with a difference of +23.96 percentage points and interval $[+7.05, +38.72]$. The primary result uses the documented valid replacement; the sensitivity preserves the original activation policy without excluding a scene.

Table 2: Paired outcomes across all 96 independent scenes.

Paired outcome	Symbol	Scenes
Both success	a	53
Ours only	b	27
Generic only	c	3
Both failure	d	13

Table 3: Prespecified paired inference. CI bounds refer to the Ours-minus-Generic risk difference, in percentage points (pp), not to the p-value.

b	c	Exact p	Difference (pp)	95% CI low (pp)	CI high (pp)
27	3	8.43033194542e-06	25.0	8.561337	39.151123

5.2 Difficulty and the Hard condition

Figure 3 and Table 4 report the realized tier outcomes. Ours/Generic successes are 35/31 of 38 Easy scenes, 26/22 of 28 Moderate scenes, and 19/3 of 30 Hard scenes. The largest observed descriptive gap occurs in the two-wall Hard condition, with 17 Ours-only and one Generic-only success. Nevertheless, Ours also fails in 11 Hard scenes. The tier results show heterogeneity in this sample; they are not three additional significance tests or evidence that all occluded scenes are solved by relevance weighting.

5.3 Where tasks stop

Table 5 and Figure 4 aggregate the original first-terminal causes into the task stages. Budget exhaustion without a confirmed candidate accounts for 29 Generic failures and five Ours failures. Thus the observed failure distribution is consistent with the proposed emphasis on acquiring evidence needed for handoff. It is not a proof that the field alone explains every downstream difference.

Ours and Generic respectively obtain at least one confirmed candidate in 89 and 63 scenes; 88 and 63 pass the common screen and complete Ground navigation and stopping. Fresh D435 refinement and actual-arrival D_{exec} are reached in 82 and 59 scenes. Subsequent physical retrieval succeeds in 80 and 56. These stages must not be collapsed: Ours has one task completing the lift stage but failing final retention, which remains a primary failure.

The remaining first-terminal bins include observation/frame/capture failures, one Ours screen failure, D435 refinement failures, descent/closure failures and one Ours retention failure. There are no primary first-terminal navigation or refined-pregrasp failures in this cohort, not a guarantee that those stages cannot fail. More Ours tasks reach Ground execution, so comparing raw Ground failure counts does not estimate a conditional execution reliability advantage. Refinement and geometric rejection remain meaningful limits of the complete system even when aerial support has been confirmed.

5.4 Recorded qualitative example

Figure 5 shows Hard-015, selected after the results as the lowest-scheduled eligible Hard Generic-failure/Ours-success pair with retained window evidence. This is an illustration, not a preselected representative scene or an additional statistical sample. Both methods accept three windows. For the displayed per-run exact anchors, Generic still lacks sufficient ground support in 19/96 covered cells, whereas Ours lacks support in 0/88. Confirmed candidate counts evolve as $0 \to 0 \to 0$ and $0 \to 0 \to 3$,

Table 4: Tier counts and paired success-rate differences (percentage points). The primary inference pools the original 96 scenes.

Tier	N	Ours success	Generic success	Difference pp
Easy	38	35	31	10.53
Moderate	28	26	22	14.29
Hard	30	19	3	53.33

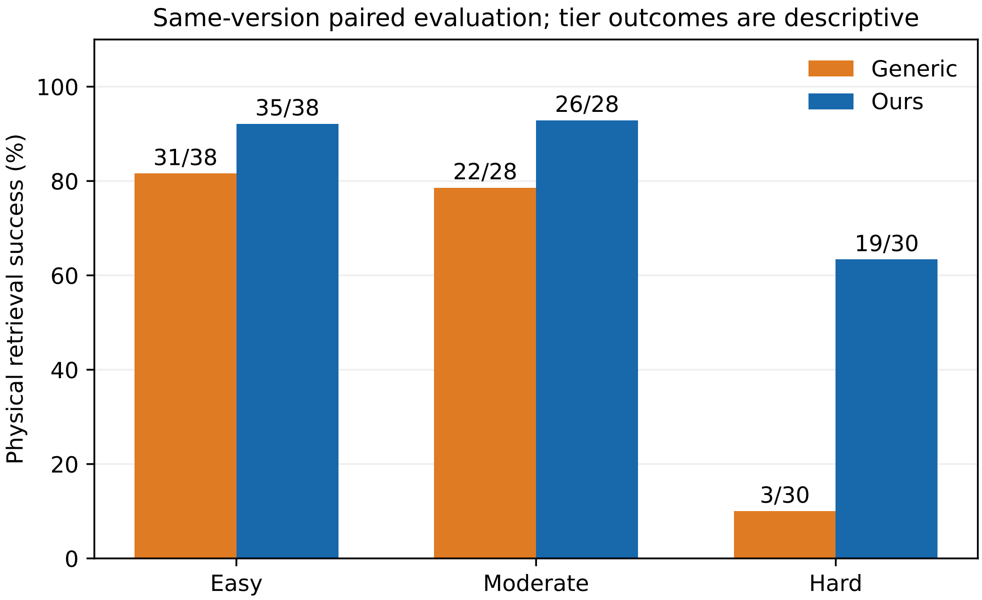

Figure 3: Descriptive retrieval success by difficulty. Denominators are the prospectively sampled 38/28/30 mixture, not an outcome-dependent balanced subset.

Table 5: Primary first-terminal failures. Unreached downstream stages are not counted as failures. Confirmation includes budget exhaustion without a confirmed candidate; observation includes early frame and capture acceptance.

Stage	Generic failures	Ours failures
UAV observation	4	2
Confirmed candidate	29	5
Screened candidate	0	1
BUNKER navigation	0	0
D435 refinement	4	6
D_exec (refined pregrasp)	0	0
Grasp (descend / close)	3	1
Lift / retention	0	1

respectively. Generic exhausts its budget without confirmation; Ours passes the shared screen after its third window and proceeds to the recorded successful physical retrieval.

The panels retain each run's native grid origin and perceived target rather than assuming identical sensor estimates or exact anchor poses. Dotted pose connections indicate observation order, not reconstructed trajectories. The ground-presence map is not a collision-free-space certificate, and the figure does not independently prove grasp contact. Its limited explanatory role is to show how different observations leave different real support deficits at handoff-relevant locations.

5.5 Observation cost and descriptive controls

On the 53 joint-success pairs, Ours uses on average 0.283 fewer accepted windows; the descriptive paired bootstrap interval for Ours minus Generic is $[-0.415, -0.150]$ windows (Table 6). The intervals for resolved location changes, active time, Ground time and whole-task time all include zero. Consequently, the data do not establish a general time or distance saving. The conditional window difference cannot be interpreted as an all-scene cost effect or as evidence of equal success rates.

Table 7 shows the scheduled auxiliary comparisons. On the six context scenes, Ours succeeds in five, Fixed-view in one and RM4D-only in two. On the four ablation scenes, Ours succeeds in two; without flight cost also succeeds in two, and without occlusion in zero. The no-cost comparison is fully concordant on this small subset. It does not establish equivalence or that cost is unnecessary. Likewise, the occlusion result is descriptive evidence from four scenes, not proof of universal component necessity. The primary isolation of task weighting remains the full Generic/Ours comparison.

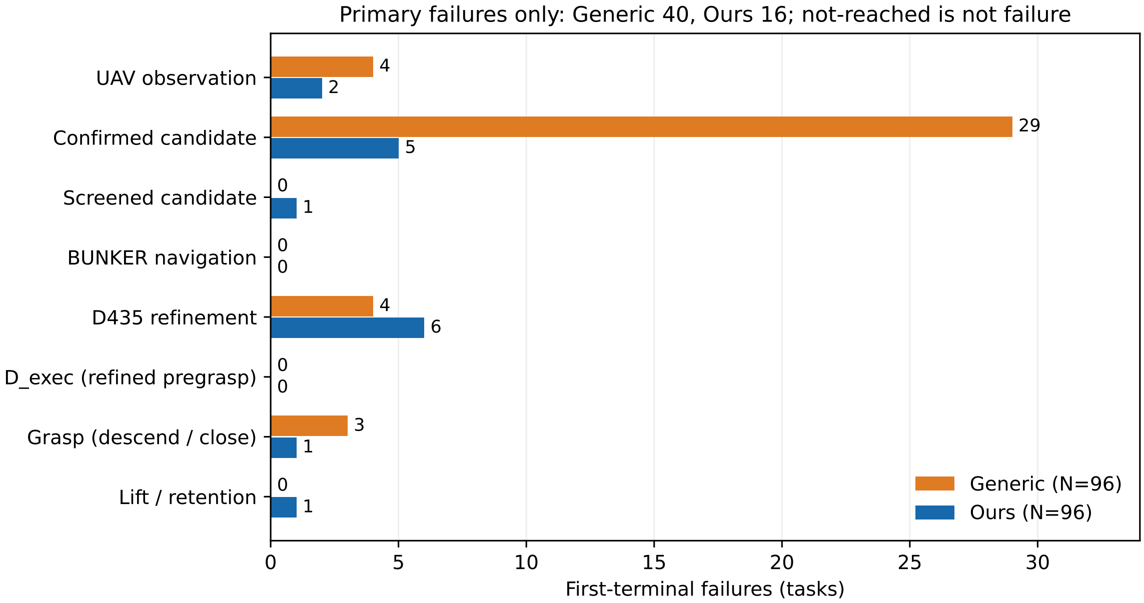

Figure 4: First-terminal distribution: 40 Generic and 16 Ours failures. These are all-task counts, not failure probabilities conditioned on reaching each stage.

Table 6: Conditional efficiency on 53 joint successes. Differences are Ours minus Generic; `T_active_sim`, `T_ground_sim` and `T_task_sim` are seconds of simulation time. Intervals are descriptive paired bootstrap intervals. Location changes are counts, not path distance.

Metric	Joint success pairs	Mean Ours minus Generic	CI low	CI high
windows	53	-0.283019	-0.415094	-0.150472
resolved_location_changes	53	-0.094340	-0.207547	0.018868
T_active_sim	53	-2.752189	-7.872662	2.804814
T_ground_sim	53	-1.518396	-7.875041	5.132873
T_task_sim	53	-8.024887	-17.004175	1.241642

6 Discussion

6.1 What the result supports

The primary evidence supports manipulation-conditioned aerial sensing for the studied retrieval population. The controlled difference is spatial weighting: the same acquisition model and candidates can direct measurements toward environment regions that support future manipulator stances rather than toward generic deficit alone. The observed reduction in no-confirmed-candidate failures is consistent with this mechanism, while the complete physical endpoint prevents candidate discovery from being mistaken for successful retrieval.

The method's relevance field should not be read as a success probability. It combines validated samples, joint-margin quality, footprint geometry and current operational evidence. It neither models all IK branches nor optimizes navigation, camera visibility and grasp execution jointly. The one-step gain holds current support fixed, although future evidence may block or reveal other poses. The remaining failures, including Generic-only successes, show why a relevance-guided policy need not dominate on every scene. No guarantee of monotonic task progress or globally optimal sensing is established.

6.2 Task, sensing and simulation limitations

The evaluated population contains one known brick, static horizontal ground, the installed robot geometry and modest perturbations of zero-, one- and two-wall layouts. Initial sensing begins in a known work region; this is not an unrestricted aerial search for arbitrary objects. The RGB-D assumptions, Ground-task reachability map, planar reference bridge and gripper geometry are task-specific. Uneven terrain, dynamic occlusion, different object geometry and unseen layouts are outside the present evidence.

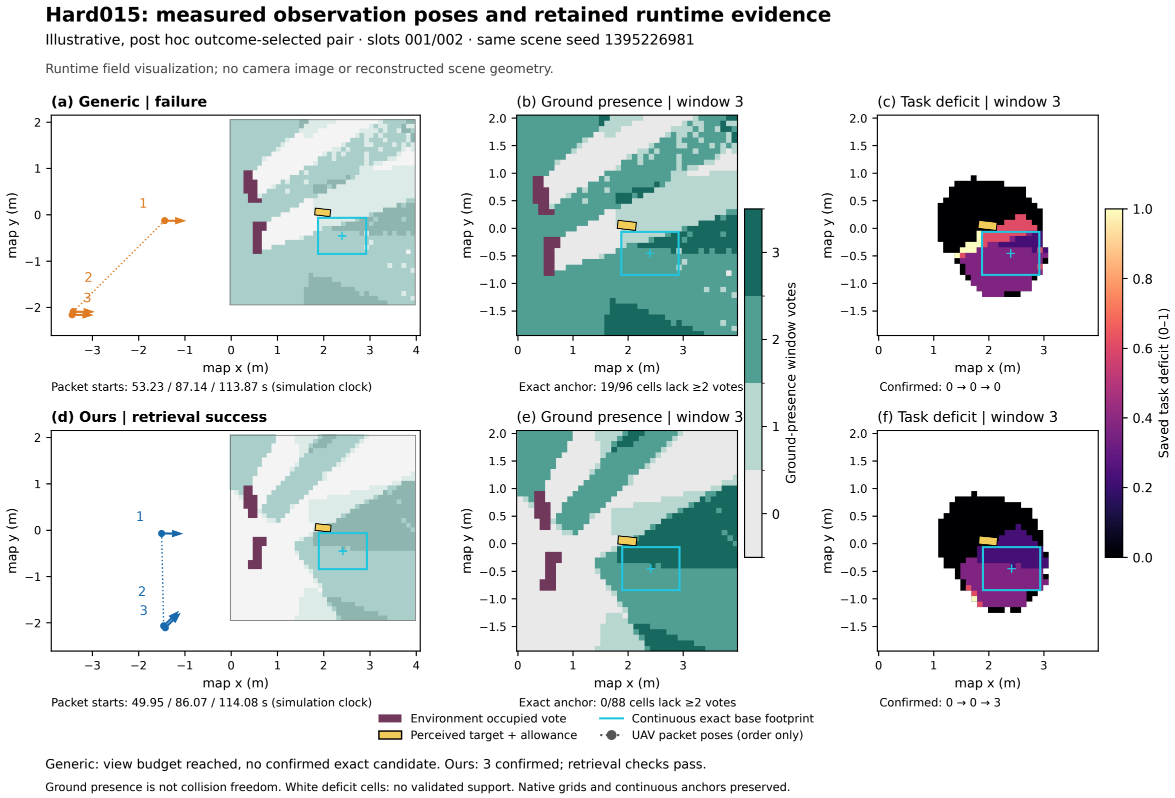

Figure 5: Post-hoc Hard-015 illustration, slots 1/2, seed 1395226981. Measured observation poses, accumulated ground-presence votes and task deficit are shown for Generic and Ours. Continuous per-run anchors are preserved. White deficit cells have no validated support; zero deficit is not a FREE label. The two anchors are not assumed identical. No scene truth or synthesized sensor image is used in these panels.

Table 7: Descriptive auxiliary comparisons using scheduled Ours counterparts. a, b, c, d have the paired meanings of Table 2, replacing Generic with the listed comparator. Subsets are not powered secondary experiments.

Comparator	N	Ours success	Comparator success	a	b	c	d
fixed	6	5	1	1	4	0	1
rm4d_only	6	5	2	2	3	0	1
no_cost	4	2	2	2	0	0	2
no_occlusion	4	2	0	0	2	0	2

Finite-scan opportunity averages a deterministic schedule over starting phases at a nominal hover pose. It does not calibrate real return likelihood, reflectance, pose error or intracloud motion distortion. Assumed-height environment occluders and optimistic unknown-space visibility can mispredict what the next window actually observes. Environmental evidence also retains coarse conservative blockers, and finite candidate validation can omit useful stances. Missing ground support remains a legitimate reason to stop within budget; predicted opportunities cannot substitute for measurements.

Physics-based execution is stronger evidence than a kinematic reachability count, but is not real-robot validation. In the current platform, public UAV map alignment uses a private simulated body-state backend, Ground global alignment is spawn-derived, and grasp confirmation uses simulated bilateral contact and joint feedback. The policy receives runtime sensor observations and public transforms, but the complete localization/contact stack is not sensor-only in the real-world sense. An independent checker uses simulated object height to verify lifting. Short retention is additionally supported by the task's grasp-confirmation hold; the study does not provide an independent continuous object-height trace throughout a long hold. Dynamics, friction, calibration and feedback differences limit transfer.

6.3 Experimental interpretation and transfer

The 96 paired scenes permit the prespecified primary comparison, not a claim of universal generalization. Difficulty is a sampled mixture, the qualitative case is outcome-selected, and auxiliary subsets are small. Conditional efficiency includes only joint successes, and incomplete distance recordings preclude a distance-saving claim. The paired success result should therefore remain the central empirical conclusion, rather than being expanded into unmeasured energy, speed or real-world robustness benefits.

Future transfer should first establish shared real map/time conventions, camera/LiDAR extrinsics and hand-eye calibration, then bind the public flight, base and trajectory interfaces to measured hardware feedback. Real grip and object-retention evidence require separate characterization; controller completion alone is insufficient. A stationary sensing/feedback dataset and offline geometry checks are a suitable first step before bounded motion tests. No real adapter or hardware experiment is included here. The current simulation versions and results remain fixed references for any later transfer study.

7 Conclusion

We presented a manipulation-aware active-perception framework in which a ground mobile manipulator's validated stance support guides an aerial robot's observations. Exact footprint projection and separate operational/ground evidence connect manipulation relevance to a finite-window sensing objective, while a shared execution interface tests whether the acquired information supports a physical retrieval. In 96 independent paired simulated scenes, Ours achieves 80 retrievals compared with 56 for the matched Generic policy, with a prespecified paired difference of 25 percentage points. The evidence supports the proposed task-conditioned allocation of sensing in this bounded setting, while leaving general efficiency gains and real-robot transfer unresolved.

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